

Is AI actually bad for the environment or are we overreacting?

Posted by u/PuzzleheadedHeat5792 • • 0 points • 73 comments

I've been reading a lot about AI lately, and one thing that keeps coming up is its environmental impact.

On one hand, AI models (especially large ones) need massive data centers. These consume a lot of electricity, require cooling systems, and in some regions even depend on non-renewable energy. Training a single large model can use as much energy as thousands of households over time.

But on the other hand, AI is also being used to *reduce* environmental impact.

So it feels like a bit of a paradox.

AI increases energy consumption, but it can also help industries become more efficient and sustainable.

Comments

u/FARHANFREESTYLER • 2 points • 2026-03-29 19:40 UTC

It's definitely not an overreaction, but it is a complex trade-off. We're in a phase where the 'compute cost' is visible, but the efficiency gains are still scaling up.

The paradox is real: you can use AI to optimize a smart grid and save gigatons of carbon, but the training process itself has a massive footprint. What's interesting is how some specialized developers are pivoting to solve this. For instance, Beetroot (<https://beetroot.co/>) has been integrating Green Ops into their custom AI development.

As a company with Swedish-Ukrainian roots, they emphasize building smaller, more efficient models rather than just throwing massive compute power at every problem. I hope the future isn't about less AI, but about leaner and more responsible AI architecture.

u/blimpyway • 2 points • 2026-03-26 06:26 UTC

Going full Pluribus would help the environment, but that's a bit overreacting.

u/ArjunSreedhar • 2 points • 2026-03-26 06:22 UTC

AI does increase energy use, and yes, it will affect the environment.

The issue is not just the technology. It is whether companies act responsibly before the damage grows.

There are options:

better cooling systems
recycled water
cleaner energy
more efficient infrastructure

But this needs real commitment, not just ROI thinking. Governments also need clear rules.
This is one area where reacting late would be a disaster.

Because this is not about a minor side effect.
It touches a basic human need: WATER.

u/dermfloork • 2 points • 2026-03-26 01:42 UTC

not related to the environment but i just realized data centers are for data

u/geneticxyz • 2 points • 2026-03-26 01:11 UTC

I don't think it's overreacting. I've been thinking and reading a lot about this recently.

I can't find any meaningful reductions in emissions or increase in water efficiency tied to AI. I hear these usually as vague future promises, mostly marketing/pitch by leading AI advocates. I've seen examples of AI used to monitor or predict environmental impact, but at the moment these seem to be small scale efforts with minimal impact. There are very few hard examples and documented benefits so far, in spite of the billions being spent (or sought out) on investment and the huge incentive to showcase any positive impacts.

On the other hand, I've seen a number of well-documented examples of negative environmental impact: globally, nationally, and locally (I'm in the US/PNW). Here's a quick grab bag:

A little bit of googling shows that in 2023, nearly 8% of *summertime* water consumption in Virginia's "Data Center Alley" on the outskirts of the DC/Baltimore metro area was to help cool water centers. Demand is expected to quadruple over the next decade.

I've read multiple reports of communities in Mexico, Chile and Paraguay facing water rationing/shortages because priority for potable water was given to cool data centers owned by US and Chinese corporations.

US coal plants that were going to be retired are being kept online to meet the demand of AI data centers. According to a recent US DOE report, "by 2028, AI alone could consume as much electricity annually as 22% of all US households."

Recently, the people in my city fought an effort by the local power company to clear cut protected forest in order to meet the power needs of giant data centers.

Is this water recycled, though? For human consumption in any way? Are you aware?

u/geneticxyz • 2 points • 2026-03-26 15:44 UTC

Human consumption. The data centers in the articles I read wanted potable/filtered water and tapped directly into the same municipal water systems that provide it to people for drinking and daily use. Also, some 80-90% of data centers that use water cooling do open system, so the water is lost through evaporation, rather than cycled back through the data center (as in closed systems).

u/PuzzleheadedHeat5792 • 1 points • 2026-03-27 03:30 UTC

Ohhh, got it. Thanks!

u/SoftResetMode15 • 2 points • 2026-03-26 00:48 UTC

i don't think it's overreacting, but it's also not as simple as ai = bad for the environment, it really comes down to how and where it's being used and how your team approaches it, for example if you're using ai to draft routine member emails or internal faqs, you're likely replacing small bits of manual work rather than creating huge new demand, but training large models and running them at scale is a different story and does carry real energy costs, so the more practical approach i've seen is to treat ai like any other operational decision, start with clear use cases where it reduces busywork or improves consistency, and be mindful about when you actually need heavy usage versus lighter tools, one thing i'd suggest is thinking about your own use, are you using ai for specific tasks that replace something else, or just adding it on top, and either way it's worth a quick review step so you're not overusing it without realizing the tradeoffs

u/Substantial-Cost-429 • 2 points • 2026-03-26 00:44 UTC

the paradox framing is actually the right way to think abt it but the answer probaly depends a lot on which AI use cases win out

the honest answer is training runs for frontier models are genuinely pretty bad. the numbers for training a single large model are wild and most of it is coming from regions with carbon heavy grids. inference at scale isnt much better

but the offset argument has real teeth in specific domains. AI optimized grid management, smarter routing in logistics, better climate modeling, materials discovery for solar and batteries. these are areas where the efficiency gains could plausibly be significant enough to matter

the thing that makes me more concerned than the energy use itself is the water consumption for cooling. that one gets way less attention but some of the western US datacenter buildouts are in places already dealing with serious water scarcity and that feels like a slow building problem

net net i think the env impact question is genuinely still open and it depends almost entirely on whether AI gets deployed in ways that actually reduce emissions elsewhere or just adds to consumption without meaningful offsets

u/Inside_Telephone_610 • 2 points • 2026-03-25 20:24 UTC

The numbers which were going around on water usage are greatly greatly overestimated. Also i dont think the water is really wasted, it doesnt just vanish out of existance. Also the AI is not really being used to reduce environmental impact that much.

u/PuzzleheadedHeat5792 • 1 points • 2026-03-26 03:17 UTC

Ohh, okay!

u/DarthEvader42069 • 2 points • 2026-03-25 19:23 UTC

It certainly does have some negative impacts, but I'd say relative to its utility, they are far smaller than the impacts of other industries. I mean just think about how much environmental harm grass lawns do. Probably at two orders of magnitude more than AI.

u/paulcaplan • 2 points • 2026-03-25 18:08 UTC

The number one thing you can do to reduce carbon footprint by far is flying less. One cross country flight is more carbon than your Claude max usage for a year.

u/PuzzleheadedHeat5792 • 1 points • 2026-03-26 03:16 UTC

Agreed!!

u/UpperYoghurt3978 • 2 points • 2026-03-25 15:16 UTC

Yes in variety of ways mostly due to pollution from fossil fuels and infrasound.

u/[deleted] • 1 points • 2026-03-25 14:09 UTC

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